

Indira Gandhi National Open University

इंदिरा गांधी राष्ट्रीय मुक्त विश्वविद्यालय



Enrolment No : 2554605810

Name : Rajeev Kumar

Address : Chandani Chauk ,11 No Lakh Buxar
Bihar

Program Title : BSCFMT – Bachelor of Science (Mathematics)

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Assignment No : 06

Study Centre Code : 07189

Phone No. : 6202720356

Email ID : krrajeev939@gmail.com

Date : 14-04-2026

Signature : Rajeev Kumar

DATA ANALYSIS

Tutor Marked Assignments

Course Code: BECS-184

Assignment Code: Asst /TMA /2025-26

Total Marks: 100

Assignment One

Answer the following questions. Each question carries 20 marks

$$2 \times 20 = 40$$

1. The following table presents the list of agencies owning different number of rental cars. The annual income for each of these agencies is given here.

Agency	Cars (in ten thousand)	Revenue (in Rs Crores)
A	63.0	7.0
B	29.0	3.9
C	20.8	2.1
D	19.1	2.8
E	13.4	1.4
F	8.5	1.5

- (a) The study wanted to analyse whether there is any significant relationship between the number of rental cars that the agency owns and its annual income. Compute the correlation coefficient from the given table.
- (b) Test the significance of the correlation coefficient considering $\alpha=0.05$ and critical values obtained from t-distribution table is ± 2.776
2. Explain the term 'qualitative research design'. How it is different from quantitative research design? What are the various techniques of qualitative research design?

Assignment Two

Answer the following questions. Each question carries 12 marks.

5 X 12=60

3. Differentiate between Census and Survey data. What are the various stages involved in planning and organizing the censuses and surveys?
4. Explain the following:
 - a. Multivariate analysis
 - b. Coefficient of variation
 - c. Data mining
 - d. Normal distribution curve
5. What are the characteristics of a good questionnaire? Suppose you have to conduct a study in village to assess the effectiveness of a recent government scheme. Prepare a questionnaire with atleast 10 questions to conduct this study.
6. What do you mean by the term 'Composite index'? State with illustration the procedure to work out composite index by simple ranking method.
7. Differentiate between:
 - a. t test and f test
 - b. Phenomenology and Ethnography
 - c. Point estimate and interval estimate
 - d. ANOVA and MANOVA

Q1 -

The following table presents the list of agencies owning different number of Rental cars. The annual income for each of these agencies is given here.

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A	63.0	7.0
B	29.0	3.9
C	20.8	2.1
D	19.1	2.8
E	13.14	1.4
F	8.5	1.5

(a) The study want to analyze whether there is any significant relationship between the number of rental cars that the agency owns and its annual income. Compute the correlation coefficient from the given table.

(b) Test the significance of correlation considering $\alpha = 0.05$ and critical values obtained from t-distribution table is ± 2.776 .

Ans - To analyze the relationship between the number of rental cars and annual income of the agencies, we will compute the person correlation and test its significance using the given data.

(a) Computation of the correlation coefficient (r):

Let :

- XXX : Number of cars (in ten thousand)
- YYY : Revenue in (Rs crores)

Agency	X (Cars)	Y (Revenue)	$\cdot X^2$	$\cdot Y^2$	$\cdot XY$
A	63.0	7.0	3968.00	49.00	441.0
B	29.0	3.9	841.00	15.21	113.1
C	20.8	2.1	432.64	4.41	43.68
D	19.1	2.8	364.81	7.84	53.48
E	13.4	1.4	179.56	1.96	18.76
F	8.5	1.5	72.25	2.25	12.75

Now calculate the necessary summations:

- $\Sigma X = 153.8$
- $\Sigma Y = 18.7$
- $\Sigma X^2 = 5859.26$
- $\Sigma Y^2 = 80.67$
- $\Sigma XY = 682.77$
- $n = 6$ (number of observations)

Use the formula of Pearson correlation coefficient:

$$r = \frac{n \Sigma XY - (\Sigma X)(\Sigma Y)}{\sqrt{[n \Sigma X^2 - (\Sigma X)^2][n \Sigma Y^2 - (\Sigma Y)^2]}}$$

Substitute the values:-

$$r = \frac{6(682.77) - (153.8)(18.7)}{\sqrt{[6(5859.26) - (153.8)^2][6(80.67) - (18.7)^2]}}$$

$$r = \frac{4096.62 - 2876.06}{\sqrt{[35155.56 - 23654.44][484.02 - 349.69]}}$$

$$r = \frac{1220.56}{\sqrt{11451.12 \times 134.33}} = \frac{1220.56}{\sqrt{1532404.10}} = \frac{1220.56}{1238.88}$$

$$r = 0.982$$

(b) Test the Significance of correlation coefficient:

We test the null hypothesis $H_0: r = 0$ against the alternative $H_1: r \neq 0$

Use the t -test formula:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Substitute the values:

$$t = \frac{0.9852 \times \sqrt{6-2}}{\sqrt{1-(0.9852)^2}} = \frac{0.9852 \times 2}{\sqrt{1-0.9706}} = \frac{1.9704}{\sqrt{0.0294}} = \frac{1.9704}{0.1714}$$

$$t \approx 11.5$$

Degrees of freedom = $n-2 = 4$

Given critical value, $t_{0.05,4} = \pm 2.776$

Since $|t| = 11.5 > 2.776$, we reject the null hypothesis.

Conclusion:-

The computed correlation coefficient is approximately 0.985, which shows a very strong positive relationship between the number of cars owned by an agency and its annual income. The t -test confirms that this correlation is statistically significant at 5% level. Therefore we conclude that there is a significant and strong linear relationship between the size of a car rental fleet and corresponding revenue of the agency. This indicates that the higher number of rental cars, greater is revenue generated and vice versa.

Q 2. Explain the term 'qualitative research design'. How it is different from quantitative research design? What are various techniques of qualitative research design?

ANS. - Qualitative research design

Qualitative research design is a systematic approach used to understand human behaviour, experiences, social contexts, and cultural phenomena through in-depth investigation and interpretation. Unlike statistics or numerical analysis, it focuses on gathering non-numerical data such as words, narratives, and observations to explore underlying meanings, emotions, and motivation. The aim is not to quantify data but to gain insights into people's perspectives and how they interpret their environment.

In qualitative research, the researcher plays a central role in data collection and interpretation. The process is often flexible, iterative and adaptive to the context of the study. This approach is especially useful in early stages of research when the problem is not clearly defined or when exploring new areas where limited prior research exists.

This design is generally used in fields like sociology, psychology, education, anthropology, media studies, and public health, especially when the goal is to explore complex social processes, lived experiences, and individual narratives in depth.

Differences between Qualitative and Quantitative Research Design.

<u>Aspect</u>	<u>Qualitative Research Design</u>	<u>Quantitative research</u>
<u>Nature of Data</u>	Non-numerical (text, audio, video, images)	Numerical (Statistical, measurable)
<u>Objective</u>	To understand meanings, experience and interpretation	To quantify variable and test hypotheses
<u>Research question</u>	Open-ended	Closed-ended
<u>Data Collection method</u>	Interview, focus groups, observations, content analysis	Survey, experiments, structured questionnaire
<u>Sample Size</u>	Small and purposive	Large and random
<u>Data analysis</u>	Thematic analysis, content analysis, narrative analysis	Statistical techniques such as regression etc.
<u>Outcome</u>	In-depth understanding and theory building	Statistical validation and theory testing
<u>Researcher's role</u>	Active participant involved in interpretation	Objective observer: maintain distance from data
<u>Generalizability</u>	Context specific, not easily generalizable	Generalizable to larger population.

In essence qualitative research design seeks depth, context and complexity while quantitative research design seeks breadth, measurement and generalization.

Techniques of Qualitative Research Design

1. In-depth Interviews:

This is one of most common techniques where researcher one-on-one interview with participants to gather detailed information about their thoughts, feelings, and experiences. Interviews are usually semi-structured, allowing flexibility.

to probe deeper into responses.

2. Focus Group Discussions:

This involves small group discussions (usually 6-12 people) led by a moderator. It is useful to exploring group dynamics, collective viewpoints and social norms. The interactions among participants often simulates new ideas and perspectives.

3. Participant Observation:

Here, the researcher immerses themselves in a setting and observes the behaviour and interaction of participants. It is widely used in anthropological and ethnographic studies and help in understanding social practises and cultural norms.

4. Case Study Method:

This technique focuses on the single case or a small number of cases in great depth. The case can be an individual, group, organization, or event. It helps in gaining a comprehensive understanding of complex phenomena in real-life contexts.

5. Ethnography:

Ethnography involves extended observation of social practises and interaction in a particular cultural or social setting. The researcher often lives within the community to get firsthand experience of their daily life and customs.

6. Content analysis:

This involves analyzing texts, documents, media content or archival material to interpret patterns, themes and meanings. It can be applied to interviews, newspapers, films, or any communicative material.

7. Narrative Analysis :

In this method, the researcher analyzes stories or personal narratives shared by individuals to understand how people construct and make sense of their experiences. It is commonly used in psychology, health studies, and education.

8. Grounded Theory :

This is both a method and theoretical approach where images inductively from data. The researcher collects and analyzes data simultaneously, modifying their approach as new insights develop.

In conclusion, qualitative research design provides a powerful tool to understand the complexity of human behaviour, especially in contexts where subjective experience and social interactions are central. It complements quantitative research by offering insights that numbers alone cannot provide.

3. Differentiate between Census and Survey data. What are the various steps involved in planning and organizing the censuses and surveys?

ANS:- Difference between Census and Survey data

Census and Surveys are both ~~are~~ important methods of data collection but differ significantly in scope and applicants. A Census is a process of collecting data from every unit of the population. It aims to complete enumeration, leaving no one out. For example, the Population Census in India is conducted every ten years, covering every individual and household. In contrast a survey collect data from a representative sample of population. It is quicker, less expensive and suitable when time and resource are limited. Surveys are commonly used in health, education, and marketing and survey. While census provides highly accurate and comprehensive data, surveys offer timely insights but may include sampling errors.

Stages involved in Planning and Organizing Census and Surveys:

1. Defining objectives: The first steps is to clearly outline the aim and scope of the census or surveys. For instance Population characteristics, employment status, or health indicators.
2. Designing the Methodology: This includes deciding whether to conduct a full census or a sample survey, choosing data collection methods and defining sampling techniques.

3. Questionnaire Design : Questions are prepared keeping the objective in mind. They should be simple, clear, and relevant.
4. Pre-testing and Pilot survey : A small-scale trial run is conducted to test the questionnaires and logic-logistics and make necessary changes.
5. Training of Investigators : Enumerators or Investigators are trained to ensure uniform understanding and collection of data.
6. Data Collection : Fieldwork is conducted as per the plan using printed or digital forms.
7. Data Processing and Analysis : Collected data is cleaned, coded, entered, and analyzed using statistical tools.
8. Report writing and Dissemination : Results are compiled, interpreted, and published for public and institutional use.

Both census and surveys play a vital role in policy making and socio-economic planning in India.

4. Explain the following :

a) Multivariate analysis

ANS:- Multivariate analysis is a statistical technique used to examine data that involves multiple variables simultaneously. It helps in understanding the relationship between different variables and how they affect one another. For example, if a researcher wants to study the effect of education, income, and age on spending habits, all three variables are considered together using multivariate techniques. Common methods include multiple regression analysis, factor analysis, and cluster analysis. These methods are especially useful in social sciences, marketing, and economics where data is complex and interrelated. It provides deeper insights and help in better decision-making by revealing patterns that might not be visible through univariate or bivariate analysis.

b) Coefficient of Variation

ANS:- The Coefficient of variation (CV) is a statistical measure that shows the extent of variability in relation to mean of the population. It is calculated as the ratio of the standard deviation to mean and is usually expressed as a percentage. $CV = (\text{Standard deviation} / \text{mean})$ and is usually expressed from one data series to another, even if the means are dramatically different. For example, in comparing income distributions in two different regions of India, CV helps determine which region has more relative variability, regardless of their average income.

c) Data mining

ANS:- Data mining refers to process of discovering patterns, trends, and useful information from large sets of data using algorithm and statistical techniques. It is widely used in field like banking, retail, healthcare, and telecommunications. For instance online retailers use data mining to predict customers buying behaviour and recommended products. It helps in decision-making, detecting fraud, and improving customer relationship. It involves step such as data collection, cleaning, analysis and interpretation to derive meaning results from raw data.

d) Normal distribution curve

ANS:- The Normal distribution curve, also known as bell-shaped curve, is a symmetric probability distribution where most observation cluster around central peak and probabilities for values taper off equally in both direction. It is defined by its mean and standard deviation. Many natural phenomena such as height, weight and test score follows a normal distribution. In data analysis, it is a crucial because many statistical test and procedure assume data to be normally distributed. This helps in making accurate inferences and predictions.

5. What are the characteristics of a good questionnaire? Suppose you have to conduct a study in village to assess the effectiveness of recent government-recent scheme. Prepare a questionnaire of at least 10 questions to conduct the study.

ANS. - Characteristics of Good Questionnaire :

A good questionnaire is a vital tool in data collection. It should be well structured, clear and relevant to research objective.

The following are some key characteristics:

1. **Clarity and Simplicity:** The language used should be simple, unambiguous and easily understood by respondents of different literacy level.
2. **Relevance:** All questions should relate directly to topic of study and contribute meaningfully to data required.
3. **Logical sequence:** Questions should be arranged in a logical order, generally starting from general to specific.
4. **Objectivity:** Questions should be free from bias and should not lead the respondent towards a particular answer.
5. **Brevity:** It should be concise to avoid respondent fatigue but should cover all necessary aspects.
6. **Close-Ended and Open-Ended questions:** A mix of both types of questions helps collect both quantitative or qualitative data.

7. Pre-tested : The questionnaire should be pre-tested on a small group to check clarity and effectiveness.

8. Culturally Appropriate : The question should be sensitive to local custom, beliefs, and language.

Sample questionnaires for studying Effectiveness of Government scheme in village.

1. Are you aware of recent government scheme introduced in your village? (Yes/No)
2. How did you know come to know about scheme? (Gram Sabha / TV / Radio)
3. Have you or your family benefitted from scheme? (Yes/No)
4. What kind of benefits you have received? (financial / Health / Other)
5. What was the application process easy to understand and follow? (yes/No)
6. How long did it take to receive the benefits after applying? (Less than 1 months / 1-3 months / More than 3 months)
7. Are you satisfied with support provided under scheme? (Yes/No)
8. What changes challenges did you face in availing the scheme? (Open-ended)
9. In your opinion, how could the scheme be improved? (Open-ended)
10. Would you recommended this scheme to other in your village? (Yes/No)

3. Ranking the Units - Rank each unit for every individual indicator. If a higher value is better, the highest value gets rank 1. If a lower value is better, the lowest value gets rank 1.
4. Summation of Ranks: Add ranks of each unit across all indicators.
5. Final ranking: The unit with lowest total rank is considered the best overall and is assigned the composite rank 1. The others follow in ascending order of total ranks.

Illustration:

Suppose we have three states (A, B and C) and two indicators literacy rate and mortality rate.

State	Literacy rate (%)	Rank	Infant Mortality	Rank	total Rank
A	85	1	30	2	3
B	70	3	25	1	4
C	75	2	35	3	5

Composite Index Rank:

- State A: Rank 1 (Lowest total rank)
- State B: Rank 2
- State C: Rank 3

This method gives a simple but effective composite measure using rank-based aggregation.

6. What do you mean by term Composite index? State with illustration the procedure to work out composite index by simple ranking method.

ANS.- The term composite Index refers to a single numerical measure that combines multiple individual indicators or variable into one index number. It helps to summarize and evaluate overall performance, status, or trends across different dimensions. This method is widely used in socio-economic studies, development planning, education ranking, health indices etc. The Composite index simplifies complex, multidimensional data into a form that is easier to interpret and compare.

Procedure to work out composite Index by Simple Ranking Method :

The Simple Ranking method is one of basic techniques for constructing a composite Index. It involves making the performances of units (like states, districts or individual) across several indicator and then combining these ranks to get a final composite rank. The steps are as follows.

1. Selection of Indicators : Identify the relevant variables or indicators to be included in the Index. for example, literacy rate, infant mortality rate, per capita income, etc.
2. Data Collection : Collect data for each unit (e.g. states) for all selected indicators.

7. Differentiate between

a) t test and f test

ANS:- The t-test and f-test are both statistical tests but are used to compare the means of two groups and determine if they are significantly different from each other. It is commonly applied in small sizes and tests hypothesis about population means.

There are different types of t-test such as one sample, independent two sample and paired sample t-tests.

On the other hand, an F test is used to compare the variances of two or more groups. It is often used in context of ANOVA to assess whether the means of multiple groups are equal by comparing their variances. It is also used to test overall significance of regression model.

b) Phenomenology and Ethnography

ANS:- Phenomenology and Ethnography are both qualitative research methodologies, but they differ in focus and approach. Phenomenology aims to understand and describe individuals lived experiences of a particular phenomenon. It is rooted in philosophy and focuses on subjective perception and consciousness.

In contrast, ethnography rooted in anthropology and aims to study the culture, behaviour, and social interaction of people in their natural setting. It involves long-term fieldwork and participant observation. While phenomenology explores "what" people experience, ethnography seeks to understand "how" people live within cultural context.

c) Point estimate and interval estimate

ANS - A point estimate provides a single value as an estimate of a population parameter. It is simple, but may not capture the true value due to sample variability.

An interval estimate, such as confidence interval, provides a range within which the parameter is expected to lie. It is more informative reliability.

d) ANOVA and MANOVA

ANS - ANOVA (Analysis of Variance) is used when comparing the means of more than two groups based on a single dependent variable. It identifies whether at least one group mean is statistically

MANOVA (Multivariate analysis of Variance), on the other hand, is used when there are two or more dependent variables. It evaluates whether the mean difference among groups on a combination of dependent variables.